

Dear Editor,

We are pleased to submit the manuscript entitled **“Detector-Plane Causality in Quantum Measurement: A Benchmark Framework for Measurement Integrity and Control”** for consideration as a Regular Article in *Review of Scientific Instruments*.

This work introduces an instrumentation-focused framework for quantum measurement systems by formalizing detector pipelines as causally ordered components. We present the Quantum Measurement Stack (QMS), a diagnostic abstraction that localizes irreversible information loss at the detector plane, encompassing sensor materials, readout electronics, digitization, firmware, and software reconstruction. Using this framework, we define QMCTB-01, a reproducible detector-plane benchmark, and demonstrate its utility through paraxial wave-optics simulations of a near-field double-slit configuration under identical photon statistics.

The results show that interference visibility is a hardware-constrained property determined by detector-plane coherence preservation rather than statistical accumulation. This establishes a practical benchmark for evaluating and controlling measurement integrity across detector technologies.

The manuscript emphasizes benchmarking methodology, detector evaluation, and reproducible diagnostics in quantum optics instrumentation—topics central to the scope of *Review of Scientific Instruments*.

All simulation code and benchmark artifacts are publicly available

GitHub: <https://github.com/srikarr20/quantum-measurement-stack>;

Zenodo: <https://doi.org/10.5281/zenodo.17966568>)

in accordance with FAIR data management principles.

This manuscript is original, has not been published previously, and is not under consideration elsewhere.

Thank you for your consideration. We look forward to your feedback and would be pleased to address any questions from the editors or referees.

Sincerely,
Srikar R
Independent Researcher